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The future of personalized learning with AI[☆]

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A B S T R A C T

Personalized learning has the potential to enhance learning outcomes but faces challenges such as problems measuring learning variables, the lack of theoretical guidance for interventions based on these variables, and ethical considerations. The availability of new AI tools, including generative AI based on large language models (LLMs), has resulted in claims that the problem of personalization has been solved. The goals of this paper are to provide a shared, systematic language for describing adaptivity, adaptability and personalization of learning environments; to broaden how we conceptualize personalization; to define the adaptation cycle; and to discuss which aspects of the cycle can be supported by AI. Also discussed are levels of AI integration, and a proposed architecture for AI-based personalized learning environments.

Education relevance: The personalization of learning, i.e., the accommodation of every person's individual needs during the learning process, has been a goal for educators and learning environment designers alike. However, achieving this goal has been a challenge as key aspects of the personalization cycle face challenges, such as the measurement of learning variables and the lack of theoretical guidance for interventions based on these variables. We also need to consider ethical considerations related to unintended consequences of personalization. In this paper, we aim to deepen the discourse about personalized learning by defining key terms, showing how to broaden current conceptualizations of personalization to include affect, motivation, and socio-cultural variables, and providing terminology for the adaptivity cycle that allows for a more conceptual clarity. We then discuss how AI can support personalization of learning, with a focus on applications beyond chatbots. We introduce levels of AI integration in personalization and discuss an architecture that may enable personalization of learning through generative AI.

1. Introduction

The personalization of learning, i.e., the accommodation of every person's individual needs, preferences, and abilities during the learning process, has been frequently described as an appealing alternative to the standardized and uniform approach of the school model of the Industrial Age in which large groups of students are taught the same material at the same pace. The significance of personalizing learning is based on research showing, for example, that the effectiveness of important instructional tools such as scaffolds and scripts critically depends on how well they are personalized, i.e., how well they align with each learner's needs (Kollar et al., 2018). Fixed fading regimes, which are based on the same predetermined fading of support for all learners, produce inconsistent learning outcomes, as they often fade support either too early or too late for many learners (Chernikova et al., 2025). In response, adaptive approaches have emerged to dynamically assess learning and adjust support in real time (Aleven, 2013). Aleven et al. (2017) propose an *Adaptivity Grid*, which identifies five key learner characteristics and three different time scales of adaptations. The grid highlights that adaptivity is multifaceted, i.e., different combinations of what is adapted and when it is adapted can support personalized learning.

Since the 1960s, computer-assisted learning systems such as PLATO, and, since the 1980s, intelligent tutoring systems, have been designed to adapt learning to individual learners. Today, an increasing number of commercial products claims to personalize learning, often without providing details for how the system adapts and without solid empirical evidence for the effectiveness of these systems (Plass and Pawar, 2020; Wang et al., 2024a, 2024b). The discourse on personalization is further complicated by a lack of conceptual clarity in the literature of what exactly is meant by personalization of learning and how it relates to other frequently used terms, such as adaptivity and adaptability. Our first goal in this paper is therefore to provide definitions for personalization, adaptivity, and adaptability in learning and, perhaps more importantly, to introduce key terms that define and clarify the specific components of the adaptation cycle.

The design of personalized learning systems that are able to improve learning outcomes faces a series of theoretical and practical challenges. These challenges include problems measuring the learning variables on which the personalization is based, the lack of theoretical guidance for how to respond to specific levels of these variables, and limitations to the ways and extent to which learning environments can flexibly respond to a learner's needs (Plass and Froehlich, 2025). Due to these limitations,

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commercially available personalized learning environments have a relatively narrow focus on mostly cognitive variables, especially the learner's current state of knowledge (Plass and Pawar, 2020). A second goal of our paper is therefore to encourage the broadening of the range of variables based on which systems personalize learning by discussing learner variables that go beyond the common focus on mastery of the subject matter.

The availability of increasingly powerful artificial intelligence (AI) tools, including generative AI tools based on large language models (LLMs) such as GPT, Gemini, and LLaMA, has provided new opportunities to address some of the challenges described above and to implement types of personalization that were previously not feasible. In fact, with the availability of LLM-based chatbots that can be used as standalone tools or integrated with existing learning environments, there are frequent claims that due to the personal nature of the conversations that these tools can facilitate, the problem of personalization has been solved (Kasneci et al., 2023). Some educators argue that achieving in-person personalized learning, particularly on a broad scale, which is challenging due to limitations in the teaching workforce and available educational resources, is now possible due to advancements in generative AI that enable more scalable solutions (Harry and Sayudin, 2023; Hu, 2024; van der Vorst and Jelicic, 2019). In particular, some suggest that the emergence of large multimodal foundation models, i.e., LLMs that are middleware connecting various AI tools, can provide personalization for a broad range of learning experiences, including the creation of personalized learning tools by the students themselves (Küchemann et al., 2025). The third goal for this paper is therefore to discuss and, in part, speculate on an alternative vision for the use of AI in which we adopt a human-in the-loop approach where AI may be used to support personalization for each of the steps of the personalization cycle. We conclude the paper by describing levels of AI integration in the personalization of learning and propose an architecture for AI-based personalized learning environments.

2. What is personalization of learning?

2.1. Definition of adaptivity, adaptability, and personalization

The need for personalized learning arises from the recognition that one-size-fits-all solutions are insufficient to meet the diverse needs of a broad range of learners (Dumont and Ready, 2023). In the discourse on learning that accommodates specific learner needs, the terms adaptivity, adaptability, and personalization are often used interchangeably. We will therefore provide definitions for these terms that show their similarities and differences in optimizing learner engagement and increase performance and outcomes (Plass and Froehlich, 2025).

Personalized learning is an umbrella term for learning systems that tailor instruction to accommodate an individual's specific needs, interests, and goals. It involves adjusting the learning experience based on one or more learner variables. This modification of elements of the learning material aims to optimize the learning process and outcomes. Personalization can be facilitated through human interactions, digital systems (e.g., virtual agents, intelligent tutoring systems), or through hybrid approaches (e.g., recommender systems). Personalization can be based on a wide range of variables (cognitive, affective, motivational, socio-cultural), and adjustments based on these variables are referred to as adaptations (Plass and Froehlich, 2025; Vandewaetere and Clarebout, 2014). Personalization often involves giving learners some degree of choice in how adaptations are applied, for example, by allowing for interest-based problem selection or selection of learning paths.

Adaptivity refers to a system's ability to modify the instruction as a result of decisions made based on real-time or pre-assessed learner data. These modifications are made automatically, without learner control, thereby reducing the need for self-regulation of learning. Traditionally, adaptivity has prioritized observable cognitive variables (e.g., mastery, prior knowledge, skill level) that are relatively easy to measure. Adaptivity can take place at three levels. Macro-adaptivity refers to group-level adjustments, such as track assignments, which cater to broader learner characteristics. Meso-adaptivity involves aligning educational content with individual learner traits, as seen in the differentiation of difficulty levels in MOOCs (Agrawal et al., 2015; Rizivi et al., 2022). Micro-adaptivity entails real-time, dynamic modifications tailored to the learner's immediate needs, exemplified by the capabilities of intelligent tutoring systems and other learning systems with dynamic difficulty adjustments (Lin et al., 2023; Plass et al., 2019). Adaptivity can be theory-based (e.g., aptitude-treatment interactions), data-driven (e.g., AI recommendations), or knowledge graph-based (e.g., intelligent tutoring systems) (Anderson, 2013).

Adaptability also involves the modification of instruction based on learner data but does not do so automatically. Instead, adaptability emphasizes learner agency, allowing individuals to control how their learning environment is adapted, for example, by choosing content, difficulty levels, pacing, or support tools that the system recommends based on learner data. Aiming to support self-regulation and metacognition (Chernikova et al., 2025), this approach offers learner-driven control in contrast to the system-driven control of adaptivity. This learner agency, however, may disadvantage learners with low self-regulation skills and low meta-cognitive skills unless scaffolds are provided. Adaptivity is more suitable for novice learners but may limit more knowledgeable learners' potential, while adaptability benefits more knowledgeable learners but may overwhelm novices (Dumont and Ready, 2023; Levinson et al., 2022). Table 1 summarizes the differences of the three terms.

2.2. The adaptivity cycle

Adaptivity is usually described in the form of a loop, or cycle, to emphasize the fact that it is an ongoing process. The adaptivity grid by Aleven et al. (2017) includes three such loops – a design loop, a task loop, and a step loop. Whereas the design loop involves decisions made by the learning designer based on learner data, the task and step loops are system-level adaptations of the learning task and steps within a task.

The specifics of the latter two loops are usually described in three basic steps. The first step is to diagnose relevant learner variables and generate a learner model. The second step is to select adaptations to the learning environment based on the diagnosed learner state. The third step is to change the learning experience with the aim to enable an improved learning performance. Whether or not the adaptation was able to achieve its intended goal can be determined by repeating this cycle, starting with the renewed diagnosis of the learner variables of interest. Models that have described this loop include the *four-process adaptive cycle (4PAC)*, which describes the *capture* and *analysis* of learner variables, the *selection* of how and when to adapt, and the *presentation* of the adaptation to the learner (Shute and Zapata-Rivera, 2012). More recent frameworks have expanded on the *four-process adaptive cycle* by integrating learning analytics. The *closed-loop learning framework (CLLF)*, for example, links data collection and data processing to adaptivity and personalization and describes the various sources for the collection of relevant data and data analyses opportunities, including multimodal

Table 1
Overview of key terms.

Terminology	Control	Example
Personalization	System adapts and/or learner choice	interest-based problem selection, content curation, learning path recommendation
Adaptivity	System adapts automatically	intelligent tutoring systems, dynamic difficulty adjustment; intelligent feedback mechanism
Adaptability	Learner choice based on smart options	student selects learning pathway, chooses difficulty levels, chooses types of scaffolds

approaches to measure latent learner variables (Sailer et al., 2024).

Our review of the literature has shown that while the adaptivity cycle is useful as a general description, more terminology is needed to be able to describe how a specific adaptation contributes to the personalization of learning, and on what basis the decision for the adaptation was made (Plass and Froehlich, 2025). We therefore propose a set of terms to describe the different elements of this adaptation cycle to allow for a more informed further discussion of different forms of personalization. The goal of providing these terms is not to propose a new model–our aim is to introduce language for each of the specific elements of the adaptation cycle, including the personalization source, method, target, aim, and time, see Fig. 1.

- *Personalization source*. A learner's need for personalization is identified by measuring one or more learner variable(s), to which we refer as source variable(s), and by determining whether they are within an expected range. The decision to make adaptations is based on the extent to which the actual state of the variable deviates from a goal state. Examples are the learner's current knowledge, level of cognitive load, affective state, or level of engagement. A decision to adapt could be made, for example, when the level of experienced cognitive load for a learner is too high (learner may be overwhelmed) or too low (learner may be disengaged). The personalization source is described as Capture and Select by 4PAC, and as Data Collection, Processing, and Analysis by CLLF.

- *Personalization method*. How the decision to adapt the learning environment based on the source variable(s) is made is described as the personalization method. This method is a logic model with evidentiary arguments for the adaptation that draw from theory, empirical research, or both. For example, in the case of determining that cognitive load was too low, should the system increase the difficulty of the learning materials (which would assume low load is due to the problems being too easy), or should it provide motivational elements (which assumes lack of learner interest in the topic)? The theory- or empirically based decision which intervention to choose as adaptation is made based on the personalization method. In our scenario this would likely require the measurement of other variables, such as learning performance or situational interest. The personalization method is in part described by Select and Present in 4PAC and by Educational and Psychological Theory in CLLF. The personalization method also describes the personalization target and personalization aim.
- *Personalization target*: The element(s) of the learning system that change to adapt the learning experience is described as the personalization target and is the manifestation of an intervention defined by the personalization method. In our example of adapting for low levels of cognitive load, this would be the level of difficulty of the content, or a motivational element, such as an intrinsic or extrinsic reward. Other examples for personalization targets are the level or modality of feedback or guidance, and the representation mode of information that represents the learning content (e.g., visual, verbal). The target is described by Present in 4PAC and by Adaptivity and Personalization in CLLF.
- *Personalization aim*: Another component described by the personalization method is the personalization aim, which specifies the aspect of the learning process that the adaptation target aims to influence. In the case of low cognitive load, the aim would be to increase cognitive engagement, either by providing more challenging content or by providing incentives. Other examples are supporting the selection of relevant information, inducing specific affective states conducive to learning, or reducing extraneous information presented. The personalization aim can be the same as the source but could also differ. While the aim is implicitly included in 4PAC and CLLF, it is not mentioned explicitly.
- *Personalization time*: There are different time scales on which an adaptation can take place. The personalization time describes whether the adaptation is recurring (dynamic) or one-time (static). The personalization time is informed by the personalization method but also depends on learner actions, such as completing a task or task step. While the time is implicitly included in 4PAC and CLLF, it is not mentioned explicitly.

Each element of the adaptivity cycle could be discussed for the opportunities and challenges involved. We will next focus on two of these challenges, the personalization source and target, because they illustrate how much work the field still has to accomplish to deliver the idea of a truly personalized learning environment (Plass and Pawar, 2020).

2.3. Personalization targets: Expanding options for adaptivity

We have previously compiled and classified different aspects of a learning environment that can be changed in response to the current state of the adaptivity source(s) (Plass and Pawar, 2020). Among the most common adaptations are scaffolds (Bruner, 2006; Piaget, 1964; Vogel et al., 2017; Vygotsky, 1978), conceptual progression, and difficulty progression. *Conceptual progression* describes how learners advance through related concepts within a specific domain, and *difficulty progression* how the level of difficulty of the next problem is based on performance in the current problem. Cognitive models like ACT-R (Anderson, 2013) and Knowledge Space Theory (Falmagne et al., 1990) are often used to guide the progression, with many intelligent tutoring

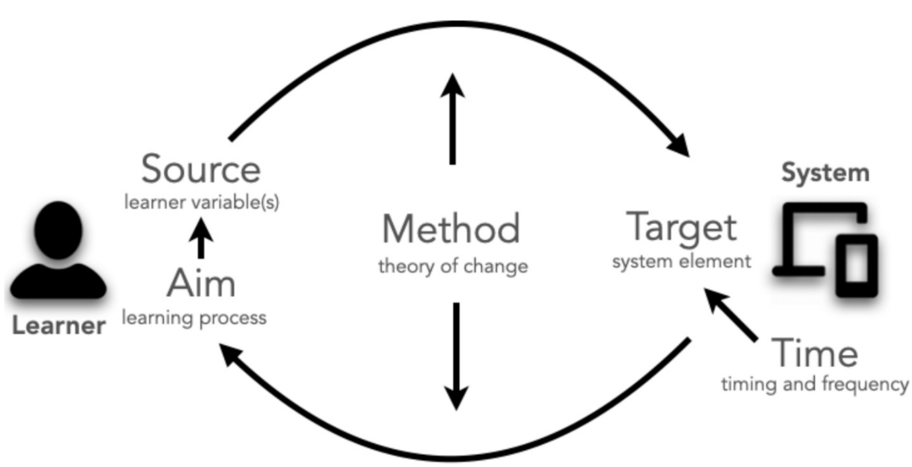

Fig. 1. Terms Describing Elements of the Adaptation Cycle and their relation to the learner and the learning environment (Plass and Froehlich, 2025).

systems relying on these models to guide learners and define learning pathways.

Less frequently adapted are other elements of the core learning activity, such as adaptations to the *interaction type*. Interaction is the reciprocal exchange between a learner and a learning environment, where the learner's reactions depend on the system's responses and vice versa (Domagk et al., 2010). Examples of interaction types include drag-and-drop, text entry, speech response, or click- or prompt-based interactions (Schwartz and Plass, 2020). The interaction type in a learning system may be adjusted based on the learner's needs, such as switching from text input to speech input when a learner struggles to type, but this is rarely done in current personalized learning environments. Similarly, the *mode of representation* describes whether information is presented visually, as on-screen text, image, video, or animation, through sound, such as speech, or through haptic channels, such as vibrations of a handheld controller. These and the other elements of the core learning are rarely used as adaptation targets, in part because few theories exist that could inform the design of corresponding adaptation methods, and in part because providing an entire learning environment's content in these various modes of representation or using different interaction types has, until recently, been prohibitively expensive.

We next turn to another element of the adaptivity cycle that has to date been implemented only in very narrow ways, which is the personalization source.

2.4. Personalization sources: Beyond cognitive variables

Most personalized learning systems use cognitive variables as personalization source, and most frequently the learner's current state of knowledge (Nakic et al., 2015). However, many other learner variables can be considered for personalization, including motivational, emotional, and socio-cultural variables, as they have been found to play significant roles in learning outcomes and learner engagement (Plass and Froehlich, 2025; Shute and Zapata-Rivera, 2012).

2.4.1. Cognitive variables as personalization sources

Personalization using cognitive variables as personalization source focuses on optimizing information processing, including how learners perceive, process, and retain knowledge. Cognitive theories like the cognitive theory of multimedia learning (Mayer, 2021) and cognitive load theory (Kalyuga and Plass, 2025) provide the foundation for these adaptations, often tailoring content delivery based on a learner's current level of knowledge, self-regulation skills, or cognitive abilities (Clark and Paivio, 1991; Leutner, 1995; Mayer, 2020). For instance, systems may adjust the difficulty of content or provide learning materials in different modalities (auditory, visual, etc.) to improve retention and comprehension. Systems such as *AutoTutor* (D'Mello et al., 2007) and *Aleks* (Falmagne et al., 1990) adjust content based on cognitive factors such as the learner's current level of knowledge. While cognitive adaptations are prevalent, other dimensions of learning, such as motivation and emotional states, are less frequently used for personalization, even though research has long established their importance as predictors of learning.

2.4.2. Motivational variables as personalization sources

Motivation plays a key role in initiating, guiding, and sustaining goal-directed behaviors (Hidi and Renninger, 2006; Ryan and Deci, 2013; Eccles and Wigfield, 2020. A learner's motivation can fluctuate throughout the learning process, and personalized learning systems can help by adjusting content or activities based on a learner's motivation level (Plass and Shao, 2026). Theories such as *Self-Determination Theory* and *Expectancy-Value Theory* highlight key elements such as autonomy, competence, and perceived value, all of which are learner variables that can inform the personalization of motivational aspects (Ryan and Deci, 2013; Stone et al., 2009). Adaptations may be made at various levels, such as by offering more engaging tasks when a learner's motivation wanes, or by framing problems in ways that are more meaningful to the learner. For example, using culturally relevant references can enhance motivation, as demonstrated by research where math problems with pop culture references led to greater engagement (Høgheim and Reber, 2015). By using adaptation to target motivation, learning systems can help maintain learner engagement, particularly during phases when a student's interest in the content may decrease.

2.4.3. Emotional variables as personalization sources

Emotions play a critical role in the learning process, influencing attention, memory, and overall engagement (Plass and Hovey, 2022; Russell, 2003) and are an important variable to consider for adaptation (Shute and Zapata-Rivera, 2012). Emotional personalization aims to manage and utilize emotions to enhance learning outcomes, such as by reducing negative emotions like frustration or anxiety and promoting emotional states that support learning. Theories such as the *Control Value Theory* and the *Integrated Cognitive-Affective Model of Learning with Media* describe emotion as individual learner variable that interacts with cognitive processes and affects learning (Pekrun, 2019; Plass and Kaplan, 2016). For example, if a system detects that a learner is frustrated, it could adjust the difficulty level, offer a break, or provide supportive feedback to alleviate negative emotions. Despite the promise of emotional personalization, challenges remain in reliably measuring emotions and determining the best adaptive strategies. The lack of a comprehensive framework for emotional adaptation in learning environments has been a barrier to more widespread implementation (Boler, 2018; D'Mello et al., 2007; Y. Kim et al., 2007).

2.4.4. Socio-cultural variables as personalization sources

Personalization of socio-cultural factors aims to increase learners' sense of belonging and relevance by situating learning in familiar, accessible contexts. This personalization is informed by *socio-cultural theory* and *situated learning theory*, which emphasize the role of social interactions and cultural context in shaping learning (Brown et al., 1989; Vygotsky, 1978). Although the integration of socio-cultural factors into learning environments is still in its infancy, some systems have begun to adapt content based on geographical or cultural preferences (Kollöffel and De Jong, 2016), such as tailoring MOOC content to learners' geo-cultural context (Rienties et al., 2022; Rizivi et al., 2022).

In summary, while cognitive adaptations are the focus of most personalized learning systems, we argue for the importance of considering motivational, emotional, and socio-cultural variables as personalization sources. We will discuss below to which extent AI may provide support for broadening the adaptation source and other elements of the adaptation cycle.

3. AI tools for the personalization of learning

There has been much enthusiasm by many scholars regarding the ability of generative AI to advance, if not solve, the challenges around personalization. Yet others argue that the use of generative AI to address personalized learning has not been adequately explored in the existing literature. While many researchers have strong positive attitudes toward the role of generative AI in personalized learning (Ellikkal and Rajamohan, 2024), others argue that there is a critical lack of cohesive analysis of generative AI's impact on learning outcomes (Guan et al., 2024; Laak and Aru, 2024). In this section we will discuss to what extent AI can, at the present time, be useful to inform the different elements of the adaptation cycle. We will begin by discussing the use of Chat Agents for personalization of learning.

3.1. Personalizing learning through chat agents

The majority of research studies published since November 2022 exploring generative AI in education focused on one type of AI application–chatbots (Lee and Hwang, 2022; Tlili et al., 2025; Wu and Yu, 2023). Chatbots are a type of AI tool that have reached widespread popularity with the availability, in 2022, of Large Language Models (LLMs) (Dwivedi et al., 2023). While earlier systems, such as ELIZA, were based on simple rules to give the illusion of meaningful conversation (Weizenbaum, 1966), LLM-based chatbots such as ChatGPT have moved toward context-aware, generative dialogue, which gave them the name *generative AI*. An LLM is an artificial intelligence system that, trained on vast amounts of text data, can both understand and generate human-like language. LLMs use layers of attention mechanisms to model relationships between words across long contexts and use learned probability distributions to generate responses (Brown et al., 2020). Users typically interact with Chatbots through on-screen text dialogues or speech. Because each conversation is based on user input, conversations are considered unique, and for that reason, chatbots are by many implicitly considered personalized by design. As is the case with most educational technologies, researchers present arguments both in favor of and against the effectiveness of personalized chatbots, with the impact on learning outcomes depending on a multitude of variables. Examples of the myriad of personalized chatbots available include *Woebot* (Prochaska et al., 2021), an AI-driven mental health support chatbot, and *Duolingo's AI chatbot* (Henry, 2023).

Some researchers claim that generative AI, including chatbots, are beneficial for students, while other studies have highlighted potential drawbacks of chatbots on learning experiences and outcomes (Bonde, 2024). For domains such as language learning, scholars found that traditional teaching methods outperformed AI chatbots, largely due to limitations in natural language understanding especially for non-native speakers, challenges in maintaining a human-like flow of dialogue, simplified feedback, input lag, and learners' unfamiliarity with chatbots (J. Kim and Lee, 2023; Mageira et al., 2022).

Studies in other domains, such as math and computer science, however, report a moderate to large effect size of the impact of chatbots on learning outcomes compared to other instruction (Laun and Wolff, 2025). Other meta-analyses argue against a mediating effect of the content domain, claiming that disciplinary differences had no significant impact on learning outcomes (Hu, 2024). Rather than the content domain, the education level and intervention time were found to mediate the effect of personalized AI chatbots on learning outcomes. Research has also shown that AI chatbots have a more pronounced impact on students in higher education compared to those in primary and secondary education (Du Boulay, 2016; Tlili et al., 2025). Furthermore, shorter intervention durations have been found to yield a stronger effect on students' learning achievements than prolonged interventions.

Applying our definitions of adaptivity and adaptability, which define these terms as providing learners with what they need in any given situation, we argue that the sole use of chatbots is a flawed way to provide a personalization of learning. Our reasoning is that a chatbot can only provide the learner with what they asked for, yet, as research has shown, learners often do not know how to ask for what they actually need (Abdelghani et al., 2024; Aleven, 2013; Zimmerman, 2000). In fact, chatbots frequently reply confidently to even poorly phrased questions, not only providing skewed responses but potentially giving learners the illusion that question-asking skills are obsolete (Laak et al., 2024).

3.2. Advanced forms of personalization with AI

While chatbots can be considered a form of personalized learning, using them for this purpose means to leave most decisions of what to personalize, and how, to the AI. We will therefore next explore other, more deliberate forms of personalization with AI, some of which are speculative in anticipation of future AI capabilities. Our goal is to contribute to a broadening of the conversation about the use of AI for the personalization of learning. As a first step in this process, we highlight that AI is not a single tool but a number of technologies and approaches that include, in addition to LLMs, diffusion models, machine learning (ML), natural language processing (NLP), computer vision, recommender systems, multimodal learning analytics, and affective computing, among others (Sailer et al., 2024; Yan et al., 2024). Each of these approaches has their own field and literature, and their detailed description is beyond the scope of this paper.

The extent to which current AI systems can effectively adapt to individual learner differences along a variety of source variables remains a critical open question. A recent meta-analysis (Hu, 2024) examined the impact of various AI-supported personalized learning systems on learning outcomes. While most educational technologies demonstrated statistically significant positive effects, one personalization method using personalized scaffolding did not yield a significant impact. In contrast, intelligent feedback mechanisms showed the largest effect size, followed by learning path recommendation systems, smart education platforms, adaptive learning systems, intelligent tutoring systems, and educational robots. These findings suggest that, while AI technologies offer promising pathways for personalization, further research is needed to understand and optimize their effectiveness of different personalization methods for different learner characteristics. Moreover, to inform future research, we believe it to be useful to deepen the discourse on personalization with AI by discussing existing and potential uses of AI for each of the elements of the personalization cycle.

3.2.1. Personalization source and the role of AI

The personalization source, i.e., the learner variable(s) on which the decision to make adaptations is based, requires two decisions. First, the learning designer has to decide which variable(s) the system should adapt for. Second, the current state of this variable needs to be measured in order to determine whether an adaptation is required at the present time. How often this measurement takes place is described by the personalization time. Generally speaking, learner trait variables can be sources for which one-time measures may be sufficient, whereas learner state variables may need to be measured frequently during the learning process.

Generative AI can assist in selecting source variables by synthesizing prior research to determine their relevance to specific learning objectives. A meta-analysis of AI supported personalized learning (Barrera Castro et al., 2024) identified five categories of source variables which were processed by AI: cognitive factors, prior knowledge, learner state, learner trait, and affective factors. The majority of studies rely on cognitive measures and prior knowledge.

Often personalization is based on one source variable, but there are examples of the use multiple learner variables as personalization sources, such as the hybrid model developed to estimate learner ability and to improve response prediction, combining Item Response Theory (IRT) and machine learning (Random Forests) (Pliakos et al., 2019). This approach relates directly to cognitive ability and prior knowledge as source variables, as it uses inferred or available prior knowledge to inform predictions of learner ability and guide adaptive content delivery.

An example using affective variables as personalization source includes a study which explored how the behavior of educational robots influences math learning with children (Konijn and Hoorn, 2020). Findings showed that all children improved, but expert learners benefited most from socially expressive robots, while novice learners performed better with robots exhibiting neutral behavior. Here, the robot's social behavior likely influenced learners' emotional engagement. A personalized learning system supported by AI would be able to decide what level of social behavior the robot should exhibit order to optimize learning outcomes.

AI language learning systems, such as *Amira* (J. J. Chen and Perez, 2023; Guan et al., 2024), can detect learning differences such as those related to dyslexia with high accuracy using rapid automatized naming. Once identified, an AI might adapt the learning experience—for example, by modifying reading fluency benchmarks, such as moving away from word count per minute as a primary measure. This demonstrates how AI can not only identify appropriate source variables but can also inform the personalization aim by adjusting which variables to emphasize based on diagnostic insights.

To effectively implement AI-supported personalization, it is critical that the value of the source variable (e.g., prior knowledge, emotion) is frequently updated. AI can contribute by facilitating in-situ measurement of source variables through ongoing learner interactions, such as through multimodal learning analytics (Ochoa et al., 2022). Examples for using AI-based measures to measure learner state variables are detecting frustration (D'Mello et al., 2007); metacognition (Azevedo et al., 2024), and motivation (Noroozi et al., 2020). However, this requires the maintenance of a dynamic learner model, which generative AI alone does not inherently support. While LLMs excel at generating personalized content and simulating feedback, their integration into systems with persistent learner modeling (e.g., through external state-tracking or hybrid architectures) is essential to accurately infer and adapt to evolving learner characteristics. Generators alone cannot measure source variables. Discriminative models—such as classification, regression, or clustering techniques—can support source variable measurement through pattern recognition and inference (Sailer et al., 2024).

Selecting and measuring the appropriate source variable comes with significant challenges. One such challenge is that learner variability on a given source variable—essential for justifying individualized interaction—is highly context-dependent, varying across learners, topics, and settings, and thus cannot be readily determined by AI alone. Moreover, the variability on most source variables increases with advanced flexible knowledge application (Chi et al., 1981; Ericsson and Smith, 1991; Larkin et al., 1980).

Lack of variability of the personalization source might become a challenge for AI models when applied in personalized learning systems because novice and expert learners most likely exhibit different levels of variance in their performance and behavior (Li et al., 2023; R. E. Wang et al., 2023). Novice learners tend to have low variance—they make similar types of mistakes or follow predictable error patterns—whereas expert learners more likely show high variance, solving problems in idiosyncratic ways. AI models trained without accounting for this unequal variance may overfit to the consistent patterns of novices and underperform when modeling experts, whose learning paths are less predictable. Personalization systems supported by AI will have to solve this challenge to be able to personalize effectively across different skill levels unless variance is explicitly modeled (Tlili et al., 2025; Zheng et al., 2023). This can lead to biased personalization, where the system is more accurate and confident for novices but provides less nuanced adaptations for advanced learners, thereby failing to support their needs.

An effective personalization method typically decides by referring to a content model, i.e., a model of what aspects of a knowledge domain a particular learner is expected to know or to acquire, a learner model including the students' characteristics, and a task model that specifies content and modality of the learning activities, none of which are part of LLMs by themselves (Park and Lee, 2008). Part of such a system is also the personalization aim, discussed next.

3.2.2. Personalization aim and the role of AI

The personalization aim, i.e., the element of the learning process that the adaptation aims to support, is usually selected by the learning designer, and may be monitored throughout the learning process. The personalization method describes how the state of the source variable(s) is used to adapt the personalization target with the goal to affect the personalization aim, which requires a learner model with latent variables. In many cases the personalization aim is the same as the source, for example, a learner's current state of knowledge. However, slow knowledge gains may not be an issue of understanding but instead be due to motivation, which would make motivation a more suitable personalization aim in some cases. LLMs can only be helpful in selecting the appropriate personalization aim in the limited cases where research on these mechanisms is available. However, learning analytics utilizing AI methods may be able to help measure aim achievement, similar to the measurement of the personalization source variable, described above.

3.2.3. Personalization method and timing and the role of AI

The personalization method consists of evidentiary arguments that describe how, based on the source variable and the adaptation aim, a specific target is adapted. Unfortunately, there is very limited empirical and theoretical guidance available for how and when the adaptation should be made, i.e., which target should be modified, how the target should be modified, and what levels of modification would have the desired effect on the personalization aim. Research designs that investigated how specific learner variables moderated the effects of specific treatments, often described as attribute-by-treatment interaction (ATI) effects, became popular in the 1960s and 1970s (Cronbach and Snow, 1977) and were used through the early 2000s (Homer and Plass, 2010; Lee et al., 2006; Plass et al., 1998), but have since been declining due to methodological reasons. Since LLMs themselves don't have a content model describing the specific domain of a topic, nor a pedagogy model, describing which learning interventions are appropriate for a specific context, they are not readily able to support the selection of a personalization method.

Given the limited research on how to adapt instruction to learner differences for many of the source variables we discussed, AI is also less suited to design the personalization method. AI tools are, however, able to assist in the discovery of new theory that can inform new personalization methods. One possibility is to use AI to identify relevant adaptation strategies by uncovering insights from related fields. For example, personalization methods may have been developed for science learning that could be identified and applied by AI to second-language acquisition. The creation of personalization methods may be supported by decision-making frameworks, such as decision trees informed by learning theories and pedagogy from a range of subjects. Another possibility is to leverage AI for data collection and learning analytics to conduct ATI research with updated methodology that overcomes the measurement and replication problems encountered in the early ATI research, such as providing better measures of the source variables and more flexible personalization targets.

3.2.4. Personalization target and the role of AI

The personalization target is the element of the learning system that changes to adapt the learning experience based on the adaptation method. AI may not be able to select the best personalization target, as this is part of the personalization method for which insufficient theoretical guidance exists, but it is well suited to generate new content or presentation modes for personalization targets. New generators for new types of media are becoming available every day, representing an area of growth and potential for the use of AI to personalize learning. The generative power of AI tools helps personalized learning to overcome one of the larges previous bottlenecks–the generation of the specific content and representation type of personalization targets. We will discuss as examples the generation of personalized scaffolding, feedback, information representations and interaction methods (Plass and Pawar, 2020).

3.2.4.1. Personalized scaffolding. An example of scaffolding involves the use of AI-generated language translations in situations when the language of instruction or assessment differs from a student's native language. In such cases, assessments—such as math tests—may inadvertently evaluate language proficiency rather than subject-specific understanding, thereby compromising construct validity. Lopez (2023) examined how dual-language scaffolds and oral/written modalities in English and Spanish, enabled Spanish-speaking English language learners to demonstrate their quantitative reasoning. The system allowed students to access items in both languages, listen to oral renditions, and respond in multiple language formats. Findings indicate that such linguistic scaffolds supported learners' comprehension, influenced response accuracy and modality choice, and significantly impacted both performance and time spent on task. Language models can play a pivotal role in offering translation scaffolds.

Another example for scaffolding is the application of AI-driven virtual agents, such as *LISSA* (Live Interactive Social Skills Assistance), in facilitating social skills training for individuals on the Autism Spectrum (Ali et al., 2020). These agents provide a safe and private environment where users engage in informal, multi-topic conversations scaffolded by real-time information on nonverbal behaviors (e.g., eye contact, facial expressions, and tone). Through multiple interactive sessions, users also receive summaries tracking skill improvement over time. The scaffolding is enabled through AI, including hidden Markov models and a schema-based dialogue manager, which automate both feedback and dialogue generation.

3.2.4.2. Personalized feedback. Another personalization target supported by AI is on-the-fly personalized feedback. Rather than relying on fixed, generic error- response mappings as in conventional systems, AI-generated feedback allows the dynamic adjustment of the type, modality, timing, and content of feedback, whereas the traditional systems simply use pre-set feedback strategies according to the type of error detected, without the ability to process instances beyond the predicted ones. An example for personalized feedback as a personalization target is Lang's study (2023) on Chinese character writing. By applying association rules and collaborative filtering, the system identifies error-prone characters and recommends feedback based on both character features and similarities between learners. The AI model leverages the successful experiences of similar students to suggest personalized stroke order corrections and pathways, effectively correcting common errors and enhancing stroke order accuracy. The approach demonstrates that mining knowledge correlations and learner similarities can significantly improve both learning efficiency and instructional effectiveness.

3.2.4.3. Personalized information representation. Generative AI can play a central role in selecting and creating content tailored to the personalization target by aligning content modalities—such as text, sound, code, or images. However, Sun and Zhou (2024) suggest that students differ in their familiarity and capacity to process various types of content: text-based content typically engages linguistic comprehension and expression, code generation emphasizes logical reasoning and functional thinking, sounds require auditory skills, while image-based content relies more heavily on visual perception and creative interpretation. These differences can translate into varying levels of academic performance depending on the content type presented (J. Kim and Lee, 2023; Sadek, 2023).

Sun and Zhou (2024) found in their meta-analysis that generative content involving text and code has a moderate positive effect size on student achievement, whereas image-based generative content shows only a small positive effect size. The authors point out that these findings may reflect both learners' cognitive preferences and the relative maturity of generative AI tools: text and code generation are often more straightforward to integrate into learning and assessment due to their logical structure and evaluability, whereas image generation can present challenges in terms of assessment standardization and technological maturity. Based on the limited empirical evidence available, generative AI is most efficient in terms of academic achievement when the personalization target is text or code-based.

3.2.4.4. Personalized interactions. Our review did not identify any studies that apply generative AI in real time to manipulate interaction types within learning environments. However, if the definition of interaction is broadened beyond conventional modes such as active, passive, click-, prompt-based and drag-and-drop interactions to include collaboration as an interactive mode, evidence exists that AI-supported personalized learning systems are more likely designed to promote independent problem-solving rather than collaborative approaches (Biswas et al., 2016; Laak et al., 2024). Face-to-face collaborative learning is rarely facilitated by personalized systems, even though AI has been used to support collaborative learning processes in some cases (Ouyang et al., 2023; Tan et al., 2022). The preference of AI-driven systems for behaviorist paradigms over social constructivist approaches may be due to the greater observability and measurability of individual learning behaviors. Implementing collaborative intelligence—the integration of human and artificial intelligence—may be more feasible in online environments, where both human and machine teaching strategies can be modeled, automated, and optimized (Tlili et al., 2025; Kabudi et al., 2021).

In conclusion, given the inherent complexity, unpredictability, and multi-agent dynamics of social learning, AI currently appears to be primarily able to support two aspects of the adaptivity cycle, which is measurement of the personalization source and aim variable(s) and the generation of personalization targets. Other elements that are not easily reducible to observable and measurable patterns remains a challenge for AI-driven personalization systems, such as determining the appropriate personalization method, i.e., the logic model describing which targets to generate for a specific state of the source variable(s) to impact a specific personalization aim. Identifying the personalization method will remain a task for human learning designers, though AI may help conduct research and theory development to provide the necessary input for the personalization method.

4. Conclusions and next steps in personalizing learning

In this section we will outline some observations we made in the current literature and discuss them as next steps. These include categories for the level of integration of AI into the personalization of learning, an architecture integrating AI agents and other components such as learner, content, and pedagogy models for personalized learning, and thoughts on avoiding unintended consequences of personalization.

4.1. Levels of AI integration in personalized learning

One issue that became apparent in our review of the literature is that the emergent field of personalized learning with AI still lacks structure. In order to categorize the use of AI for personalization, we therefore propose the following levels of AI integration in personalized learning, summarized in Table 2.

Level 0 integration is the use of LLM-based chat agents in learning environments where learners inquire about specific topics, and an LLM-based chat agent responds. This is currently the most common use of generative AI for personalized learning, but it requires learners to know what they don't know and to formulate a meaningful inquiry to the chat agent. Research has shown, however, that the design of useful LLM prompts is challenging for non-AI experts (Zamfirescu-Pereira et al., 2023). A very popular example of this level of AI integration is the use of LLMs to facilitate Socratic questioning, which has shown mixed results

Table 2
Levels of AI Integration in Personalized Learning based on number of variables considered as personalization sources and the nature and number of the personalization target.

	Personalization Source	Personalization Target
Level 0	Language (with visuals)	Language (Chat)
Level 1	Single Source Variable	Single Mode Adaptation
Level 2	Multiple Source Variables	Single Mode Adaptation
Level 3	Single Source Variable	Multimodal Adaptation
Level 4	Multiple Source Variables	Multimodal Adaptation

(Blasco and Charisi, 2024). We consider this level 0 as it largely relies on a single LLM to facilitate all aspects of the personalization without a human (other than the learner) involved.

Level 1 integration of AI bases personalization on one source variable that can be observed using one or more data streams, such as user logs and utterances, and uses a single mode of a personalization target, most frequently language, as response. As we described above, the most common variable used at this level is the learner's current knowledge in the subject matter. For example, a user's submitted coding assignment may indicate a lack of knowledge in a particular aspect of coding, and language-based feedback may be generated using AI in response, with the goal to provide the learner with hints about how to revise their code. Alternatively, the incorrect part of the code may be highlighted. Research has shown that AI-based feedback may initially result in an overreliance of learners on the LLM-generated feedback, but that learners were able to correct this over several attempts (Pankiewicz and Baker, 2023).

Level 2 integration of AI bases personalization on multiple source variables that can be observed using one or more data streams, such as user logs, facial expressions, gestures, and language, using multimodal learning analytic methods involving AI. Similar to level 1, a single mode of a target is used for the response. For example, based on a user's user logs and utterances, a state of high cognitive load may be detected. In response, the learning environment may reduce the visual complexity of the information presented, for example, by applying the chunking principle (Mayer and Moreno, 2010).

Level 3 integration of AI bases personalization on one learner variable that can be observed using one or more data streams. In contrast to level 1 integration, in level 3, responses are generated using AI to modify multiple targets in the learning environment. For example, a user's actions may indicate a lack of motivation, and the AI may generate a playful form of engagement with the learning content that consists of personalized visuals, interactions, and sound.

Level 4 bases personalization on multiple learner variables that can be observed using one or multiple data streams, such as user logs, facial expressions, gestures, and language, similar to level 2. However, unlike in level 2, responses are generated using AI to modify multiple targets in the learning environment. For example, a user's actions and physiological responses (EDA, HRV, EEG) may indicate a state of emotions not conducive to learning, and the learning environment may use AI to generate visuals and audio that are personalized to the learner to induce emotions conducive to learning.

As discussed above, LLMs by themselves are not able to support advanced forms of personalization through LLM-based chatbots alone (level 0). We therefore propose the following architecture that integrates important missing components mentioned above as an additional next step.

4.2. A RAG-enhanced multi-agent architecture for personalized learning

Our review of the literature found that some of the major shortcomings of LLMs for personalization of learning are that they do not include a learner model, a content model, and a specific pedagogy/intervention model. In order to design a personalized learning system that would overcome this shortcoming, a multi-agent architecture may be used that includes these three models, augmented by RAG (retrieval augmented generation) or similar mechanisms. This approach combines the strengths of multi-agent systems, which involve different specialized components working together, with RAG, a technique that enhances language models by allowing them to access and incorporate specific external knowledge. The high-level architecture for such a system is shown in Fig. 2.

This architecture consists of several agents that access and update models and that are enhanced by RAG.

4.2.1. Orchestrator agent

This agent manages the flow of information and coordinates the activities of the other specialized agents. It receives the initial requests or learning goals and delegates specific tasks to the appropriate specialized agents. It synthesizes the multi-agent inputs and generates the final natural language and other responses to personalize a user's learning experience.

4.2.2. Learner profile agent

This agent is responsible for creating, maintaining, and updating the Learner Model, which is based on the learner's learning history including user logs and (multimodal) measures of the personalization source. This agent may also include separate cognitive, affective, motivational, and emotional state agents. Augmentation includes theoretical and empirically based information about relevant learner variables and their measurement.

4.2.3. Content agent

This agent identifies and retrieves relevant learning materials from the Content Model and domain knowledge from external sources using its RAG capabilities. This agent may also include a separate content generator agent, which, based on the learner profile agent and the pedagogy agent, creates new learning content and modes of interactions if they are not available in the current inventory. It may also include a separate learning path agent that, in collaboration with the pedagogy agent and the learner profile agent, dynamically generates personalized learning paths. Both are augmented with domain knowledge and curricula.

4.2.4. Pedagogy agent

This agent selects and implements suitable instructional strategies from the pedagogy model based on the information in the Learner Model and the available content, augmented through subject-specific pedagogical research and best practice heuristics. This agent may also include a separate feedback agent that generates personalized feedback on learner performance, augmented by research and theory on feedback. Augmentation includes personalization methods derived from theory

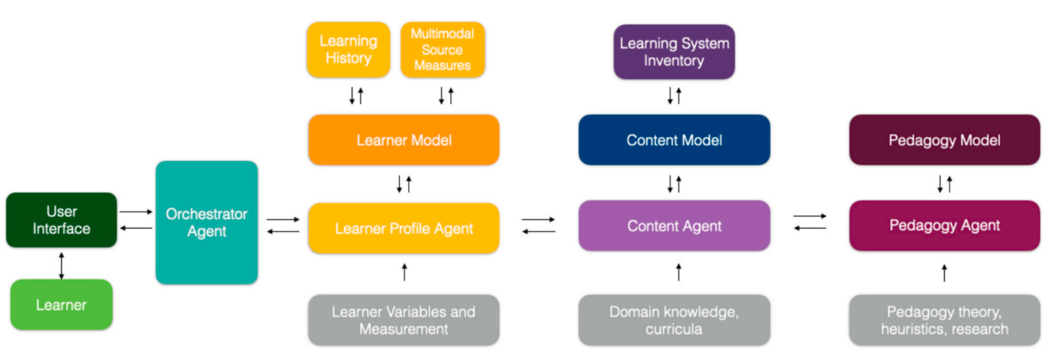

Fig. 2. Proposed RAG-enhanced Multi-Agentic Architecture for Personalized Learning.

and empirical findings along with information in which contexts they are suitable.

This architecture is not supposed to be a working system but describes the building blocks of components that are required to support personalized learning using AI.

4.3. Avoiding unintended consequences of personalization

A discussion of the use of AI tools for the personalization of learning must also include a consideration of potential downsides of personalized learning and the role of generative AI in mitigating or accelerating them. While the benefits of personalized learning—such as tailored instruction and improved learner outcomes—are frequently celebrated (Laak and Aru, 2024), it is equally important to explore the potential unintended consequences. As with any technological advancement, the implementation of personalized learning systems may produce both beneficial and detrimental outcomes. Two particularly concerning areas are the risk of increasing educational inequality and challenges involving assessment of outcomes of personalized learning (Mageira et al., 2022; Tlili et al., 2025).

4.3.1. Increasing educational inequality

While many researchers argue that personalization can help bridge achievement gaps (Roberts-Mahoney et al., 2016; Sokolowski and Ansari, 2018), this is by no means guaranteed. In fact, some forms of personalization can increase inequality. For instance, tracking (grouping students by perceived ability) may simplify teaching but tends to disadvantage those in lower tracks, reinforcing existing disparities (Terrin and Triventi, 2023). Personalized learning may also widen the test-score gap because the frequently used mastery approach results in situations where higher-performing students progress more quickly through the personalized content than lower-performing peers. Having to spend more time to complete a module for a topic means less of an opportunity to study other topics, which can result in lower scores on standardized tests (Dumont and Ready, 2023).

Further, the uncritical use of personalized learning with generative AI may undermine the goal of personalized learning by homogenizing student outputs and limiting creative and cognitive diversity (Laak et al., 2024). Instead of fostering individuality, generative AI outputs are more likely to produce similar responses across learners rather than diverse results, thereby narrowing performance variability and potentially obscuring the discovery of unique talents and learning trajectories. Research shows that, although AI tools like ChatGPT can reduce task completion time and enhance short-term performance metrics, they may also lower the quality of output, particularly among more experienced users and expert learners (Brynjolfsson et al., 2023; B. Chen et al., 2023; Dell'Acqua et al., 2023).

4.3.2. Assessment of outcomes of personalized learning

As learning becomes increasingly personalized, questions should be raised about the compatibility of personalized learning paths with standardized assessments. If assessments remain uniform while instruction is highly individualized, alignment and fairness of assessments may be compromised. A seemingly natural extension of the increased prevalence of AI-generated content in education is the use of AI for assessments. However, if both instruction and assessment are personalized through generative AI, new challenges emerge regarding the validity, reliability, and fairness of such assessments (Bondie et al., 2019). Concerns related to employing generative AI to personalize assessment include bias, hallucinations, lack of transparency, and limited control over outputs. Arslan et al. (2024) caution that these issues undermine the interpretability and trustworthiness of AI-generated evaluations. The current lack of transparency of many AI systems makes it difficult to understand or explain the rationale behind specific assessment decisions (Bender et al., 2021; Kasneci et al., 2023; Ye et al., 2023). In other words, while standardized assessments may not be compatible with personalized learning, it is not yet clear when personalized assessments will be ready to replace them.

5. Limitations

This paper aimed to provide a shared, systematic language for describing personalized learning environments and the methods of personalization they employ. As this contribution is primarily conceptual and integrative rather than empirical, some limitations arise from the way the literature was identified and synthesized. The selection of sources was guided by relevance to the key concepts of personalization, adaptivity, and AI-supported learning, with an emphasis on recent publications in educational technology and learning sciences. While this approach allowed for a focused and coherent discussion, it was not conducted as a systematic or scoping review, and therefore some relevant studies—particularly those from different educational contexts or non-English-speaking regions—may not be represented.

The proposed terminology and architecture reflect an effort to establish a common language for describing and comparing personalized learning environments, but further empirical work is needed to validate and refine these concepts in practice. Finally, theoretical boundaries remain: the framework is grounded in current models of adaptivity and AI integration and may require revision as these technologies and related theories evolve.

6. Conclusion

AI tools, including, but not limited to generative AI, hold substantial promise for the support of personalized learning. We discussed new possibilities AI offers across five personalization dimensions: source, target, aim, method and timing. The most significant opportunities are for AI to be used to generate personalization targets, i.e., the part of the learning environment that is modified in response to the learner's needs, characteristics, or preferences. We discussed examples of generating personalized scaffolding, feedback, information representations and interaction methods, while recognizing that there are many other targets to consider for personalization.

Another significant contribution of AI to the personalization of learning is the measurement of personalization source and aim variables, which must be both relevant to learning outcomes and measurable within the system. AI can process these variables through learning analytics techniques such as item response theory combined with machine learning or affective modeling, such as emotional design influencing learners' emotions, thereby broadening source and aim beyond cognitive variables.

The personalization method, which describes the logic model for the intervention based on the source variables, requires human intervention since not enough theoretical guidance is available. Relatedly, the selection of the personalization aim and personalization targets, which are both aspects of the personalization method, largely remains a human-led instructional design task that needs to be supported by theory and empirical findings.

For AI tools to be used to personalize learning for a range of different contexts and applications, they require a learner model, content model, and pedagogy model. We proposed a blueprint of an architecture for a related RAG-enhanced multi-agent system. However, successful personalization requires more than just technological solutions. It involves fostering pathways rooted in human agency, contextual understanding, and pedagogical insight—dimensions that may be hindered rather than enhanced by unchecked AI use. Therefore, while AI offers potential as a supportive tool, it must be carefully integrated into educational ecosystems with humans in the loop to avoid diluting the very heterogeneity that personalized learning seeks to promote.

CRediT authorship contribution statement

Jan L. Plass: Writing – original draft, Supervision, Project administration, Conceptualization. **Fabian Froehlich:** Writing – original draft.

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